

Spike-Driven-Transformer+Spik4lite Demo

Introduction

This is a demo implementation of "**Spik4lite: Refactoring Neuromorphic Sparsity for Efficient Spiking Neural Networks on Commodity Edge Devices**". The demo follows the **train-and-infer** pipeline to show the basic usage of Spike-Driven-Transformer+Spik4lite.

Directory

The **Spike-Driven-Transformer** is the root and its structure is described as below.

```
Spike-Driven-Transformer/  
├── README.md  
├── cifar10  
│   ├── cifar10.yml  
│   ├── test.py  
│   ├── train.log  
│   └── train.py  
├── cifar10-dvs  
│   ├── CIFAR10DVS.yml  
│   ├── test.py  
│   ├── train.log  
│   └── train.py  
├── cifar100  
│   ├── cifar100.yml  
│   ├── SOPs_consumption_on_cifar100.py  
│   ├── test.py  
│   └── train.py  
├── dvs-gesture  
│   ├── autoaugment.py  
│   ├── SDT Power profile.csv  
│   ├── SDT+Spik4lite Power profile.csv  
│   ├── SOPs_consumption_on_dvs-gesture.py  
│   ├── test.py  
│   ├── train.py  
│   └── utils.py  
├── dvs_utils  
├── model  
├── module  
└── criterion.py
```

- * **README.md** provides the general overview and introduction of the project, as well as instructions on how to run the code.

Running the Demo

The whole pipeline contains three major steps: (1) activate the Conda Environment, (2) Run the train code and Run the SOPs consumption code.

(1) activate the Conda Environment

```
cd Spike-Driven-Transformer  
conda activate Spik4lite
```

(2) Run the code

Runing on CIFAR10

Setting hyper-parameters in cifar10.yml

```
python cifar10/train.py
```

Runing on CIFAR100

Setting hyper-parameters in cifar100.yml

```
python cifar100/train.py
```

Runing on CIFAR10-DVS

Setting hyper-parameters in CIFAR10DVS.yml

```
python cifar10-dvs/train.py
```

Runing on DVS128 Gesture

```
python dvs-gesture/train.py
```

Runing on SOPs consumption

```
python SOPs_consumption_on_dvs-gesture.py
```